

CS 3015 - Robotics

Assignment I – Requirement Document

Faculty: Umarah Qaseem

You are required to **design** a robotic arm with the following workspace and configuration space specifications. The robotic arm will have a certain number of joints and links. You are free to use as many degrees of freedom as you want according to the requirements. While designing this robot, keep in mind not to make a redundant configuration space (1 or maximum 2 EXTRA joints are fine for this assignment). After designing the robotic arm, you are required to simulate it in MATLAB, apply forward and inverse kinematics.

Code:

Complete running code of your project including a driver code which will take inputs from user and call the following functions. Make a menu driven program if you want to give user choice to run all these options. (Remember marking for this assignment is relevant based on efforts)

Required Functions in your code:

Forward Kinematics function1: Input will be the values of all joint coordinates and output should be a pose (transformation matrix).

Forward Kinematics function2: Input will be the values of all joint coordinates and output should be a vector (value of x, y and z i.e. the position of end effector in 3D space). Describe singular points if any in your configurations.

Inverse Kinematics function: You are required to write a function for inverse kinematic. Input to function: Cartesian coordinates to any point, and output will be the joint coordinates required to reach that point i.e., function will take any configuration/pose of end effector and find out all joint coordinates for that pose.

GUI:

GUI is not required but marking will be relevant based on efforts. Therefore, if GUI is present credit will be given.

Plots and Visualization:

- 1- Plot results of forward kinematics by running it in a loop and plotting all the poses/positions of end effector that can occur. (This loop will take time to process and plot, do not worry).
- 2- Plot your robot link structure using the dh table.
- 3- Any other plots that you can think of.

Handwritten:

- 1- Diagram of your robot, with link lengths, thetas and ds.
- 2- Frame assignment diagram of your robot. (Do not assign frame on diagram 1, make a separate diagram.)
- 3- DH table with reasons for using this particular configuration of your robot along with limitations of your coordinates if any.
- 4- Any other schematics diagram of material that you think can help understand the configuration of your robot.

Report:

A word document (converted to pdf later) in proper report format. Learn how to present your efforts! It should contain everything written above in a structured way. Complete Code should be present at the very end of your report. Although you can put screen shots of a function or two which is unique or if you want to highlight something. The handwritten requirements should be submitted by hand, but they should also be scanned and be present in report. Write any functions or features that you have added in report as well.

Configuration Space:

You can use any combination of revolute and prismatic joints. You can use as many numbers of joints as you want. Although you will have to justify your

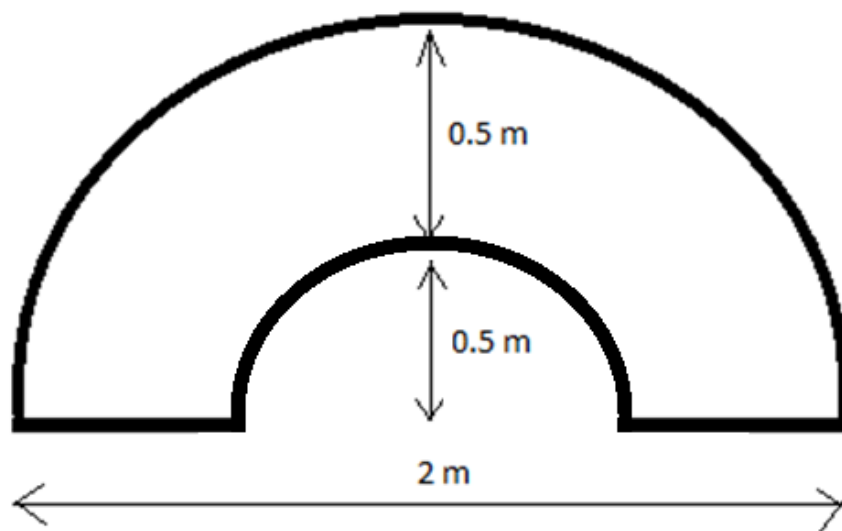
configuration space and every joint that you will add in your arm. So, think carefully while designing and do not make your space redundant.

Workspace:

**It is understood that above functions should only operate inside workspace.
(Hint: You need validation checks for this)**

The shape of your workspace is half donut. The half donut is not hollow, it is solid. The end effector of robotic arm should reach every point in that half donut space. It should not reach any point outside that space.

Workspace Dimension:



Important:

Implement your project step by step in the following sequence. This way if you ever miss out on any part, or if you find any part difficult than the other, then you will at least get the marks of what you have implemented. The following sequence is already following difficulty level in ascending order. Please follow it strictly.

- 1- A schematic diagram of your robotic arm with dh table on paper.
- 2- DH table in code.
- 3- Forward Kinematics
- 4- Plots and Visualization
- 5- Inverse Kinematics

Try to keep your code modular by writing code in functions.

Effort counts most! Marking is relevant.

Deliverables:

- 1- MATLAB simulation Code file (Name it as Rollnumber_firstname_lastname_code)
- 2- Report (Name it as Rollnumber_firstname_lastname_report)
Put both these in a zip file and submit on GCR. Name it as (Rollnumber_firstname_lastname_assignment1). Handwritten material will be submitted in class the next day.

Evaluation:

- Effort counts most! Marking is relevant.
- Through Code, Report and in the form of demo & Viva

Group Members:

This is individual assignment.

Plagiarism:

**STRICT POLICY ON CHEATING THROUGH ANY SOURCE.
ZERO MARKS WILL BE ASSIGNED IN ALL ASSIGNMENTS AND
QUIZZES.**

Announcement Date:

25th September 2024

Deadline:

13th October 2024 11:50 PM (After Extension)

This date will not be extended any further.